



Standardizing Data

A standard score tells us how far a data value is from the mean.

Converting Raw Scores into z-Scores

Antibiotics are strong medicines that are prescribed by a doctor and often used to treat infections. Once a patient begins taking an antibiotic, the length of time that it takes to cure the infection can vary.

Suppose that the time it takes a certain antibiotic to cure an infection is known to be normally distributed with a mean of 5.2 days and a standard deviation of 1.1 days.

To get an idea of how effective the new antibiotic is with certain patients, actual results can be converted into *z*-scores. A *z*-score tells us the number of standard deviations that a data value is from the mean of the data set.

When a data value (called a raw score) is greater than the mean, its corresponding *z*-score will be positive. When a data value is less than the mean, its corresponding *z*-score will be negative.

Converting Raw Scores into z-Scores

A **z-score** is the number of standard deviations that a data value is from the mean. Every data value in a data set has a corresponding *z*-score and is obtained by the formula

$$z = \frac{x - \mu}{\sigma}$$

where *z* is the *z*-score, *x* is the raw data value, μ is the mean of the data set, and σ is the standard deviation.

► **Remember** z-scores are sometimes called standard scores.

Suppose the antibiotic cures an infection in a patient in 3.8 days. What is the corresponding z-score?

Here, $x = 3.8$, $\mu = 5.2$, and $\sigma = 1.1$. These values can be substituted into the formula to find the corresponding z-score.

$$\begin{aligned} z &= \frac{3.8 - 5.2}{1.1} \\ &\approx -1.3 \end{aligned}$$

A z-score of -1.3 means that the antibiotic cures the infection in a time that is 1.3 standard deviations below the mean.

Converting z-Scores into Raw Scores

Suppose a patient takes the antibiotic and is cured of the infection according to a z-score of 2.3. This means that the patient was cured in an amount of time that is about 2.3 standard deviations greater than the mean cure time.

But suppose you want to know how many days it took the antibiotic to cure the infection in this patient. In other words, what is the raw score if the z-score is 2.3?

The z-score formula can be rearranged to create a formula for converting z-scores back into raw scores.

Converting z-Scores into Raw Scores

A z-score can be converted back into its original value, called a **raw score**, using the formula

$$x = z \cdot \sigma + \mu$$

where x is the raw score, z is the z-score, μ is the mean of the data set, and σ is the standard deviation.

► **Remember** A raw score refers to the original data value.

Now convert the z -score of 2.3 back into a raw score. Using $z = 2.3$, $\mu = 5.2$, and $\sigma = 1.1$, you can find the value of the raw score, x .

$$\begin{aligned}x &= 2.3 \cdot (1.1) + 5.2 \\ &\approx 7.7\end{aligned}$$

In this patient, the antibiotic cured the infection in about 8 days—more than 2 days longer than the average cure time for this antibiotic.